

Logic definitions



- 1
- a The process of reasoning to a general conclusion through observation of specific cases.
 - b The process of reasoning to a specific conclusion from a general statement.
 - c A sentence that can be judged either true or false.
 - d A statement that conveys only one idea.
 - e A statement that consists of two or more simple statements.

- 2
- a Yes
 - b Yes
 - c No
 - d Yes

- 3 Example answer: Soup, pasta, and carrots.

Compound statements and negation

- 4
- a
 - i Compound statement
 - ii Conditional ($\implies$)
 - b
 - i Simple statement
 - c
 - i Compound statement
 - ii Biconditional ($\iff$)
 - d
 - i Compound statement
 - ii Negation ($\sim$)
 - e
 - i Compound statement
 - ii Conjunction ($\wedge$)
 - f
 - i Compound statement
 - ii Conjunction ($\wedge$)
 - g
 - i Compound statement
 - ii Conditional ($\implies$)
 - h
 - i Simple statement
 - i
 - i Compound statement
 - ii Biconditional ($\iff$)

5



- a Some humans are not able to swim.
- b Some houses are not connected to the internet.
- c All birds have flippers.
- d Some Libras are Capricorns.
- e Some tricycles have two wheels.
- f Some cats do not have feet.
- g No cars are driving across the bridge.
- h All doctors are paediatricians.
- i Some skydivers are scientists.
- j No biscuits contain sugar.

Symbolic form

6

a $\sim$ b $\wedge$ c $\vee$ d $\implies$ e $\iff$

7

a $p \wedge \sim q$

b Conjunction

8

a $p \implies (q \vee r)$

b Conditional

9

a $\sim (p \implies \sim q)$

b Negation

10

a $(p \implies q) \vee \sim r$

b Disjunction

11

a $(p \vee q) \implies r$

b Conditional

12

a $(p \implies q) \iff \sim r$

b Biconditional

13

a $p \iff (\sim q \vee r)$

b Biconditional

14

a $(p \wedge q) \implies r$



b Conditional

15

a $(p \iff q) \vee r$

b Disjunction

16

a $\sim p$

b $p \wedge q$

c $\sim q \vee \sim p$

d $\sim q \iff \sim p$

e $\sim p \implies \sim q$

f $\sim q \wedge p$

17

a $\sim q \iff p$

b $p \implies \sim q$

c $\sim p \wedge q$

d $\sim p \wedge \sim q$

e $\sim (p \vee q)$

f $\sim (\sim q \implies p)$

18

a James does not have a smartphone.

b Dave has a laptop and James has a smartphone.

c If Dave does not have a laptop, then James has a smartphone.

d Dave does not have a laptop or James does not have a smartphone.

e It is false that Dave has a laptop and James has a smartphone.

19

a Quentin does not have a rabbit.

b Quentin has a rabbit or Maximilian has a dog.

c Quentin does not have a rabbit if and only if Maximilian does not have a dog.

d It is false that Quentin has a rabbit or Maximilian has a dog.

e Quentin does not have a rabbit and Maximilian does not have a dog.

20

a $(p \wedge \sim q) \wedge r$

b $(\sim p \wedge q) \wedge r$

c $(p \wedge q) \vee r$

d $p \implies (q \vee \sim r)$

21

a $(r \wedge q) \implies p$

b $(\sim p \iff \sim q) \vee \sim r$

c $(r \iff q) \wedge p$

d $q \implies (p \iff r)$

22



- a The water is 28°C or the sun is down, and we do not go swimming.
- b The water is not 28°C , and the sun is down or we go swimming.
- c If we do not go swimming, then the sun is down and the water is 28°C .
- d If the sun is down then we go swimming, and the water is 28°C .
- e The sun is down if and only if the water is 28°C , and we go swimming.

23

- a The water is 60°F and the sun is up, or we go snorkelling.
- b If the sun is up then the water is 60°F , or we go snorkelling.
- c If the sun is up and we go snorkelling, then the water is 60°F .
- d If the water is not 60°F , then the sun is up or we go snorkelling.
- e If the sun is up, then the water is 60°F if and only if we go snorkelling.